

Abhishek Poudel

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EDUCATION

University of Maryland, College Park

Bachelor of Science in Computer Science, December 2025

SKILLS / COURSES

- Languages: JavaScript, Python, Java, SQL
- Frameworks & Technologies: React, Node.js, Express.js, FastAPI, MongoDB, MySQL, WebRTC, Socket.IO, REST APIs, JWT Authentication, OpenAI API, Tesseract OCR, RapidFuzz, Ollama, bge-m3, Llama 3.2, RAG, Docker, Git, Linux
- Relevant Coursework: Algorithms, Advanced Data Structures, Computer Systems, Web Application Development, Artificial Intelligence, Machine Learning, Computer & Network Security

PROJECTS

Real-Time Video Calling App | React, Node.js, Express.js, WebRTC, Socket.io

Spring 2026

- Engineered a peer-to-peer video communication platform using WebRTC and Socket.io, providing low-latency browser-to-browser audio and video streaming.
- Designed a custom Socket.io signaling server to exchange SDP offers and ICE candidates, enabling reliable peer-to-peer connection establishment.
- Integrated browser media APIs and managed connection lifecycles, improving reliability through reconnect handling and browser-to-browser communication recovery.
- Validated the platform across **2-4 concurrent peer connections**, achieving an estimated **85-90% connection success rate** with an average call **setup time of 3-5 seconds**.

AI Conversational Assistant | MERN Stack, OpenAI API, React, Node.js, Express

Spring 2026

- Built a full-stack AI chatbot using the MERN stack and OpenAI API, delivering context-aware conversational responses in approximately **1-2 seconds**.
- Architected **RESTful backend APIs** to manage chat sessions, conversation history, and OpenAI interactions to support scalable request handling.
- Developed a responsive React frontend featuring real-time message rendering, loading indicators, and API error handling for an improved conversational experience.
- Optimized API request handling and frontend rendering performance by testing **100+ prompts with feedback from 5 users**.

LectureRAG Assistant | Python, Ollama, bge-m3 Vector Embeddings, Llama 3.2

Spring 2026

- Designed a Retrieval-Augmented Generation (RAG) assistant that enables students to query lecture transcripts instead of manually searching notes or emailing instructors.
- Implemented an end-to-end semantic retrieval pipeline using **bge-m3 embeddings, cosine similarity search, and a locally hosted Llama 3.2 model through Ollama** for response generation.
- Indexed **10-15 lecture transcripts** into vector embeddings and retrieved the most relevant transcript sections in approximately **2-4 seconds** per query.
- Improved response relevance by splitting lecture content into searchable chunks and supplying the **top 3 most relevant sections** to the local LLM for answer generation.

AI-Powered Alcohol Label Verification | Python, FastAPI, React, Tesseract OCR

Summer 2026

- Built a full-stack alcohol label verification application for a U.S. Department of the Treasury/TTB take-home assessment to automate comparison of label data against expected application information.
- Developed a FastAPI validation pipeline using Tesseract OCR, regex extraction, normalization, and RapidFuzz fuzzy matching to verify **5 compliance fields**.
- Tested OCR and validation logic across tilted images, multiline warnings, unit variations, incomplete fields, and OCR misreads, with **PASS / FAIL / NEEDS REVIEW** handling for uncertain results.
- Achieved approximately **2-3 second processing time per label image** and implemented single and batch uploads through a React interface.

WORK EXPERIENCE

Montgomery College - Math & Science Tutor

August 2021 - August 2022

- Tutored 20+ students in mathematics and science through one-on-one and small-group instruction, strengthening students' problem-solving abilities and conceptual understanding.
- Adapted lesson plans and explanations to accommodate individual learning styles, improving student comprehension and confidence.
- Mentored students for 10-15 hours per week through coursework support and exam preparation in Pre-Algebra, Calculus I, Calculus II, and Physics I.